1 The recommended project

Recommendation. Analyse how recorded permit timelines vary with permit type, building age, property characteristics and estimated work value. Use the findings to guide process investigation and applicant expectations. The downloaded and integrated dataset already exceeds the required size and variable thresholds.

This choice fits the teaching approach you described: a meaningful decision, careful aggregation, understandable graphs and defensible statistical evidence. The recommendation rests on data suitability and analytical scope. It does not assume that a construction topic alone will earn a better evaluation.

The decision the analysis should support

A permitting operations team needs to decide which categories deserve closer process review and what timeline ranges to communicate. Applicants need realistic expectations for comparable work. A useful result could identify a high-volume category with a long upper tail, supported by counts and uncertainty. The records do not contain staff hours, so precise staffing requirements or savings cannot be estimated.

Options reviewed

Candidate and credible sources	Assessment for this assignment
SF permit timelines Building Inspection + Assessor-Recorder + BLS [S1-S5]	**Selected.** 154,689 unique permit rows are downloaded. Clear parcel and month joins; broad scope for distributions, grouped summaries and hypothesis testing.
NYC permit and property analysis DOB applications + City Planning PLUTO [S6-S7]	Strong alternative with substantial fields and volume. The legacy DOB application dataset excludes DOB NOW jobs, so system coverage and migration need extra work. PLUTO offers tax-lot context.
Construction inspection patterns OSHA inspections/violations + BLS QCEW [S8-S9]	Authentic, relevant safety records. Inspection selection, changing coverage and employee exposure denominators make interpretation more demanding. Inspections are not a random sample of all sites.
Indian infrastructure overruns MoSPI flash reports [S10]	Very relevant locally, but the inspected October 2023 report covers 1,788 projects. Reaching 100,000 defensible rows would require a genuine multi-period panel and substantial extraction. Repeating rows to meet the threshold would be invalid.

The shortlist was screened using official documentation. Only the selected San Francisco sources were fully downloaded, integrated and audited. The alternatives are not claimed to have passed a complete data audit.

Scope that keeps the project feasible

Study all eligible building-permit categories filed in 2019-2024, including alterations. A permit is an administrative unit, not necessarily a unique building site or a complete construction project. Restricting the work to new buildings would discard most observations and could break the 100,000-row requirement. Subgroup analyses can be smaller when their sample sizes are reported.

2 The authentic dataset

One permit per row with historical property context and monthly monetary context

Source	What was acquired and why it is useful
SF Building Inspection i98e-djp9 [S1-S2]	55 recorded fields; all 154,689 primary-address permits filed 1 January 2019 through 31 December 2024. Source dates, descriptions, statuses, costs, units and stories are retained.
SF Assessor-Recorder wv5m-vpq2 [S3]	18 recorded fields from fiscal rolls 2017-2022. Scanned 1,266,295 parcel-year rows and retained the 92,364 distinct rows used in the join. Includes built year, property area, class, stories, units and zoning.
US Bureau of Labor Statistics CUUR0000SA0 [S4-S5]	72 published monthly CPI observations for 2019-2024. Supports a transparent, broad purchasing-power adjustment of estimated work value. This is a separate federal source, not another copy of the permit data.

How the sources become one consistent table

Permit grain. Select primary_address_flag = Y, the same rule used by the publisher's primary-address view. Confirm unique permit_number and record_id. The source has one record per permit at an address; using all addresses could duplicate a permit in aggregates.

Property join. Preserve block and lot as strings, including leading zeros. Strip surrounding whitespace and normalise case. Left-join on block + lot + closed_roll_year = filing year minus 2. For a 2019 filing, use the 2017-2018 fiscal roll, which ended before filing. This conservative lag trades some freshness for clearer historical ordering; later corrections to source records remain possible.

Monthly join. Join CPI to the filing year-month. Its 72 observations are contextual values repeated across permits. They are not 154,689 independent measurements of inflation. Both joins are validated as many-to-one, and the permit count must remain unchanged.

Verified audit item	Observed result
Final dataset and duplicates	154,689 rows; 95 columns; zero duplicate permit numbers
Recorded versus calculated fields	55 permit + 18 assessor + 2 BLS fields + 20 derived/integration/QA fields
Property join	151,378 matched permits (97.86%); 3,311 unmatched retained
CPI join	100% of permit rows matched to their filing month
Issuance date quality	143,200 valid intervals; 11,471 missing issue dates; 18 negative intervals flagged
Key missingness	Estimated cost 8.70%; approval date 10.63%; valid building age unavailable for 5.80%

These counts describe the downloaded snapshot, not a future guaranteed portal count. Acquisition occurred on 23 October 2026 India time (22 October UTC). Source update timestamps, request URLs, hashes and column definitions are included in the package.

Useful variables and units

Calendar days: filing, approval, issuance and final inspection intervals. USD: estimated cost and revised estimates. Square feet: assessor property area and lot area. Categorical: permit type, existing/proposed use, construction class, neighbourhood and status. Derived fields include building age, filing month, issuance within 365 days and clearly labelled estimate revisions. No attributes or observations were simulated.

3 What the first graphs reveal

Exploratory feasibility checks from the actual integrated data

The overall median hides the mix of work. Among 143,200 permits with valid issue intervals, 55.80% were issued on the filing date. The median is therefore 0 days, while the 90th percentile is 104 days. Of these issued records, 133,108 are type 8 over-the-counter alterations. A single citywide average or median cannot describe every type of project.

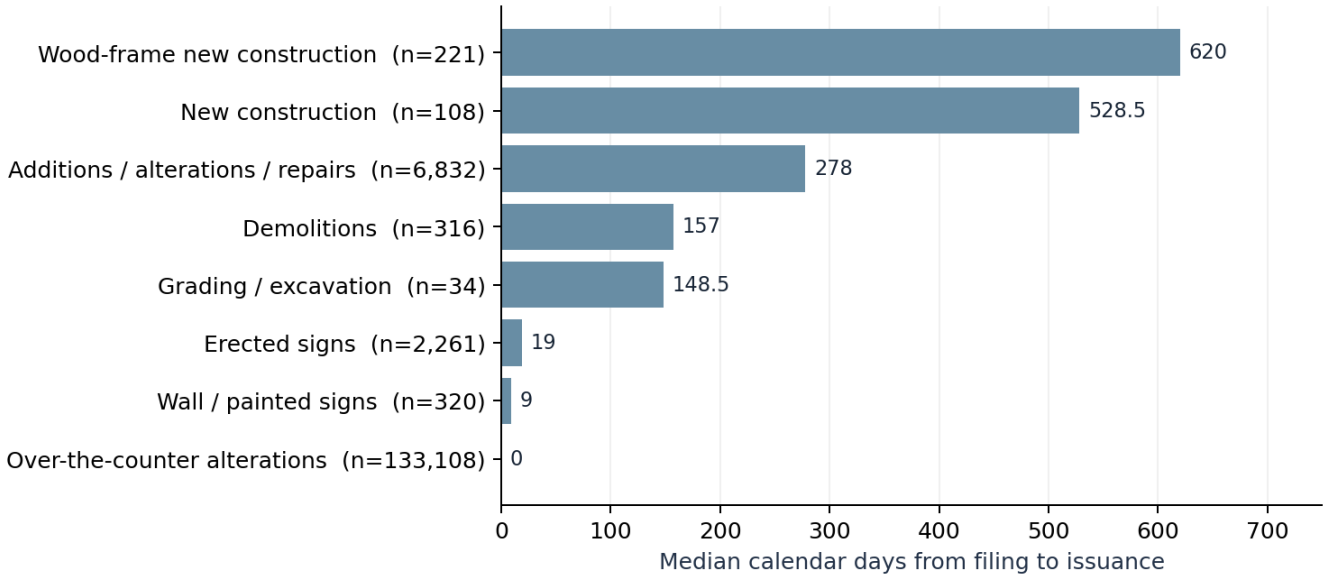


Figure 1. Medians among issued records with valid nonnegative filing-to-issuance intervals. The n labels show each group's denominator. Type 3 has a median of 278 days; type 8 has a median of 0. New-construction groups are much smaller. These are unadjusted associations, not measured departmental delay or proof of a process failure.

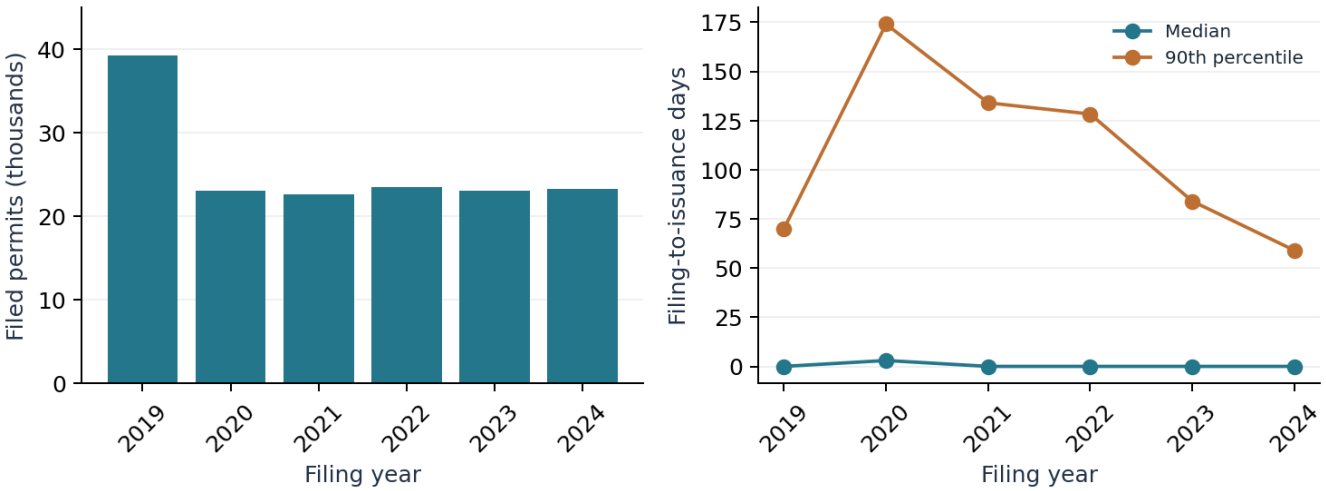


Figure 2. Filing cohorts differ in volume and their upper-tail timeline. Later cohorts have less follow-up time, and permit mix may change. Use common 365-day outcome windows and type-specific comparisons before interpreting trends. The 2020 pattern does not, by itself, identify a COVID effect.

These preliminary summaries establish that the data supports a substantive project. They are not final hypothesis tests. The package contains the underlying year/type aggregates so every number can be checked.

4 Research questions and graphs

Turn each question into a defined comparison and a decision

Question and initial hypothesis	Planned evidence and interpretation
H1 Permit categories Do issuance timelines differ across recorded permit types?	Compare median and 90th percentile, issuance within 365 days, and group sizes. Use chi-square only as a screen; base final inference on parcel-clustered resampling and confidence intervals. Repeat within filing year.
H2 Building age Are older buildings associated with longer timelines within comparable type/year groups?	Use the lagged assessor built year. Compare age bands 0-29, 30-59, 60-99 and 100+ years, with missing age separate. Apply a stratified permutation comparison or rank-based test plus clustered intervals. Adjust the comparison for property size where feasible.
H3 Estimated work value Do higher cost quartiles have longer timelines within the same type/year?	Use positive estimated costs, with zero and missing values separately reported. Compare quartiles and within-group Spearman association with valid intervals. Repeat in nominal within-year USD and CPI-adjusted USD. This is not a cost-overrun hypothesis.

For each hypothesis, state a null of no association or group difference, the eligible records and the effect measure before running the full test set. Use a 5% significance level with Holm adjustment across the three primary hypothesis tests. Report practical effect sizes and 95% confidence intervals, not p-values alone. Statistical tests may reject tiny differences in a large dataset.

The graph plan

Graph	Aggregation and question it answers
Monthly line chart	Count unique permits by filing month; reveal changing workload and seasonality.
Sorted horizontal bar chart	Show counts by permit type and neighbourhood. Use shares with denominators, not unexplained rankings.
Grouped median and percentile plot	Compare medians and 90th percentiles by type and year. Avoid averaging medians across groups.
Box plot plus a full-range view	Show issuance spread by type or age band. Use a log1p axis if needed; retain valid extremes in calculations.
Binned cost plot	Within type/year, plot median interval against estimated-cost quartile, with group size and uncertainty.
Missingness and join chart	Compare match/missingness rates by type and year. Check whether the property-linked analysis excludes particular kinds of permits.

Keep the outcome definitions explicit

The primary duration is issued_date minus filed_date, measured in calendar days among valid issued records. Complement it with issuance within 365 days among all sufficiently observed filings. A missing recorded issuance is not proof of active backlog. Report current cancelled and withdrawn categories separately; exclusions using current status are sensitivity checks, not historical status reconstruction.

5 Course methods and execution

Use BADIR as the structure and add complexity only when it helps the decision

BADIR stage	Concrete work and deliverable
Business question	Define the stakeholder decision, cohort, unit of analysis and KPIs. Success is a reliable comparison and an actionable process-review shortlist.
Analysis plan	Record H1-H3, nulls, outcome definitions, subgroup rules, tests and limitations before the full analysis.
Data collection	Use the supplied real source extracts, validate types and joins, and preserve lineage. Reproduce counts in SQL; demonstrate a Spark SQL aggregation if expected in the lab.
Insights	Generate interpretable charts, stratified comparisons, tests and robustness checks. Classify hypotheses as supported, unsupported or inconclusive.
Recommendations	Prioritise categories for process investigation; propose applicant timeline guidance and monitoring KPIs. State what further operational data is needed before staffing changes.

Mapping to the handout

Core. Decision-centric framing and KPI design; ETL and SQL aggregation; feature engineering and selection; hypothesis testing; reproducibility; bias and measurement checks. These are directly relevant to the problem. A formal experiment, propensity matching, regression discontinuity or difference-in-differences is not justified merely because it appears in the syllabus.

A limited modelling extension. If the instructor expects predictive work and a timeline estimate would help, compare a permit-type median baseline with one small gradient-boosting regressor. Fit on 2019-2022, tune on 2023, and assess on 2024. Report MAE, median absolute error and performance by permit type. The 2024 cohort has already been inspected descriptively, so do not describe it as an untouched blind test.

Use only plausibly available filing-time attributes, preprocess on the training set, and exclude issue/approval/completion dates, current status, revised cost and outcome-derived fields. The live source is not a versioned filing-time feature store; current values can have changed. Any model remains a retrospective benchmark until that availability is verified. PCA, transformers and reinforcement learning have no clear benefit here.

Production topics. Preserve source versions, schema checks and experiment records. Use MLflow if models are compared. Include deployment infrastructure only when required or useful. Confirm this focused scope with the instructor: the project guideline asks for all relevant prescribed methods.

A feasible work sequence

Milestone	Output
Already prepared	Official source acquisition, 95-column integrated snapshot, dictionary, join audit and preliminary graphs.
Next 2-3 working days	Team verifies raw examples; checks missingness, status definitions, unmatched parcels and outliers; finalises hypotheses.
Following 3-4 working days	Stratified charts, hypothesis tests, cluster-aware uncertainty and sensitivity checks; optional baseline/model comparison.
Final 2-3 working days	Decision recommendations, limitations, reproducible repository, final report and concise stage-based slides. Align dates to the instructor's deadline.

The handout requires a public GitHub repository and flags AI-generated code/plagiarism under UFM. Confirm permitted use of this preparation aid; every member should explain and reproduce the submitted work.

6 Limitations and viva preparation

Defensible claims depend on knowing what the records do and do not measure

Risk	How to handle it
Changing project mix and selection	Compare within type/year, show counts and contrast matched versus unmatched permits. Do not infer neighbourhood discrimination from raw duration differences.
Unissued and cancelled cases	Report missing outcomes and current statuses. Use common follow-up windows. Never remove all non-issued records and then claim a complete backlog analysis.
Historical and property mismatches	Use lagged rolls; disclose the 3,311 unmatched permits. Parcel splits, new construction and condominium changes can break exact joins. Historical records can be revised.
Unobserved operational causes	Applicant revisions, payment waits, staff hours and review-routing details are not fully represented. Elapsed time does not establish agency fault or the cause of delay.
Money and time definitions	Revised estimate is not actual spending. Final inspection date is not active work duration. National CPI is broad price context, not a local construction-cost index.
Outliers and inference	Flag impossible dates and report exclusions; retain plausible extremes. Cluster by parcel because one building can have many permits. A census of records still has measurement uncertainty and limited transferability.

Questions every member should be ready to answer

Why this dataset? It meets the mandatory volume, variable and source rules and supports a practical decision. **What does one row represent?** One primary-address permit, not a site. **Why left joins?** To preserve the permit population and expose missing context. **Why a lagged assessor year?** To avoid using a later fiscal period as prior property context.

Why can the median be zero? More than half of valid issued permits are same-day, largely within the dominant OTC category. **Can you claim a cause?** No; these comparisons are observational. **What if H2 is unsupported?** Report that result and uncertainty. **What has been completed?** Data preparation and feasibility checks; formal hypothesis validation remains to be done.

Sources and required submission materials

[S1] SF DBI Building Permits and metadata

[S2] Publisher primary-address view and dictionary

[S3] SF Assessor Historical Secured Property Tax Rolls

[S4] BLS CPI data and series CUUR0000SA0

[S5] BLS API signatures and CPI interpretation

[S6] NYC DOB legacy Job Application Filings

[S7] NYC City Planning PLUTO documentation

[S8] OSHA inspection and enforcement data

[S9] BLS QCEW employment and wage data

[S10] MoSPI October 2023 flash report, p 38

Assignment sources: ADS_Project_Guidelines.pdf, Advanced Data Science.pdf and ADS_Pre_Submission_Guidelines_Concise.pdf, supplied by the student and read in full. The pre-submission requires one printed page and all eight specified items. Its viva schedule is 23 October 2026 for ADS3/4 and 26 October for ADS1/2, during the scheduled lab. Fill the group number, all names and student IDs before printing.

Source documentation checked on 23 October 2026 IST. The provided metadata files contain exact acquisition timestamps, full API request URLs and raw source definitions. All reported feasibility statistics were computed from the delivered snapshot.